Here is a document carefully detailing the Experimental Data creation process

Our aim is to have a perfectly orthogonal matrix where the off-centered diagonals are approximately 0.

Step 1. True Model Coefficients

```{r true model coefficients}
set.seed(123)


# ---- True Model Coefficients ----
beta0 <- 650000
beta_size <- 5000
beta_reno <- 150000
beta_urban <- 200000
beta_dist <- -2000
beta_interaction <- 2000
```

Here we specify the true model, this is not the perfectly true model from the first part because we removed one variable(bed), due to strong correlation with size

Why we dropped bed

- Impossible to balance without breaking correlation
- Breaking correlation affects the true model
- Size already captures the information bed would capture because of the direct correlation, **bigger house = more beds.**

Understanding our Coefficients

- **beta0 = 650,000** — baseline house price (intercept)
- **beta_size = 5,000** — each additional m² adds R5,000
- **beta_reno = 150,000** — a renovated house is worth R150,000 more
- **beta_urban = 200,000** — urban location adds R200,000
- **beta_dist = −2,000** — each extra km from the city reduces price by R2,000
- **beta_interaction = 2,000** — the size effect is R2,000 stronger per m² in urban areas

Step 2. Create Perfect Orthogonal Grid

```{r Create Perfect Orthogonal grid}
# ---- Create Fully Crossed Orthogonal Design ----
design <- expand.grid(
  size_cat = c("Small", "Medium", "Large"),
  dist_cat = c("Near", "Far"),
  renovated = c("No", "Yes"),
  location = c("Rural", "Urban")
)

# Replicate each cell 90 times
design <- design[rep(1:nrow(design), each = 90), ]
rownames(design) <- NULL

# Verify dimensions and balance
dim(design)
table(design$size_cat, design$dist_cat)
table(design$renovated, design$location)
```

This part of the code creates every possible combination of the four categorical variables, we get 24 unique cells and we replicate them 90 times to get 2160 total obs. We do this because if every combination of factor levels has exactly equal representations, the design matrix is orthogonal, meaning all predictors are completely uncorrelated with one and other eliminating multicollinearity by construction.

Step 3. Add Continuous Variables

```{r Add Continuous Variables}
# Size - continuous within each size category
design$size <- ifelse(design$size_cat == "Small",
                      runif(nrow(design), 50, 131.59),
               ifelse(design$size_cat == "Medium",
                      runif(nrow(design), 131.59, 212.8),
                      runif(nrow(design), 212.8, 300)))

# Distance - continuous within each dist category and location
design$dist_km <- ifelse(design$location == "Urban" & design$dist_cat == "Near",
                         runif(nrow(design), 1, 16.75),
                  ifelse(design$location == "Urban" & design$dist_cat == "Far",
                         runif(nrow(design), 16.75, 32.5),
                  ifelse(design$location == "Rural" & design$dist_cat == "Near",
                         runif(nrow(design), 32.5, 66.25),
                         runif(nrow(design), 66.25, 100))))

# Check
head(design)
summary(design$size)
summary(design$dist_km)
```

The previous step only creates a grid, now we inject continuous variables into the cells, this chunk assigns continuous values drawn from uniform distributions within the boundaries of each category.

Step 4. Simulate the Response Variable

```{r Response Variable}
# ---- Simulate Response Variable ----

# Error term
epsilon <- rnorm(nrow(design), mean = 0, sd = 50000)

# Simulate price using true model
design$price <- beta0 +
                beta_size * design$size +
                beta_reno * (design$renovated == "Yes") +
                beta_urban * (design$location == "Urban") +
                beta_dist * design$dist_km +
                beta_interaction * (design$size * (design$location == "Urban")) +
                epsilon

# Check
summary(design$price)
head(design$price)
```

Here we generate house prices using the true model, the error term adds realistic noise representing unmeasured confounders, all the coefficients are explained in step 1.

Step 5. Fit the Model

```{r Linear Regression Model}
model <- lm(price ~ size + dist_km + renovated + location + size:location,
            data = design)
summary(model)
```

Here we just fit the model the more interesting part is the result

```
Residuals:
    Min      1Q  Median      3Q     Max 
-170628  -33334    -456   32747  171644 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)        637963.32    6301.14  101.25   <2e-16 ***
size                 5038.65      21.47  234.71   <2e-16 ***
dist_km             -1930.02      70.62  -27.33   <2e-16 ***
renovatedYes       152679.82    2167.85   70.43   <2e-16 ***
locationUrban      207159.57    6713.32   30.86   <2e-16 ***
size:locationUrban   1967.94      30.42   64.69   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 50370 on 2154 degrees of freedom
Multiple R-squared:  0.9916,	Adjusted R-squared:  0.9916 
F-statistic: 5.077e+04 on 5 and 2154 DF,  p-value: < 2.2e-16
```

Linear Regression Analysis Report

1. Model Overview

A multiple linear regression model was fitted to examine the determinants of property prices. The explanatory variables included property size, distance from a reference point (in km), renovation status, location type (urban vs non-urban), and an interaction term between size and urban location.

2. Model Fit

R-squared: 0.9916

Adjusted R-squared: 0.9916

Residual Standard Error: 50,370

F-statistic p-value: < 2.2e-16

Approximately 99.16% of the variation in property prices is explained by the variables included in the model. The overall regression is highly statistically significant.

3. Interpretation of Coefficients

- Size:

For non-urban properties, each additional unit increase in size increases price by approximately 5,038.65.

- Distance (km):

Each additional kilometre from the reference location decreases price by approximately 1,930.02.

- Renovation Status:

Renovated properties are valued at approximately 152,679.82 more than non-renovated properties, holding other variables constant.

- Urban Location:

Urban properties have a base premium of approximately 207,159.57 compared to non-urban properties.

- Interaction (Size × Urban):

The interaction term (1,967.94) indicates that the effect of size is stronger in urban areas. For urban properties, the marginal effect of size is 7,006.59 per unit.

4. Residual Analysis

The median residual is close to zero (-456), indicating no systematic bias in prediction. Residuals range between -170,628 and 171,644, suggesting some variability in extreme cases but generally stable model performance.

5. Conclusion

Property size is the primary driver of price, with urban location significantly amplifying the value of additional space. Renovation status adds a substantial premium, while greater distance reduces property value. Overall, the model provides a highly accurate explanation of property price variation.

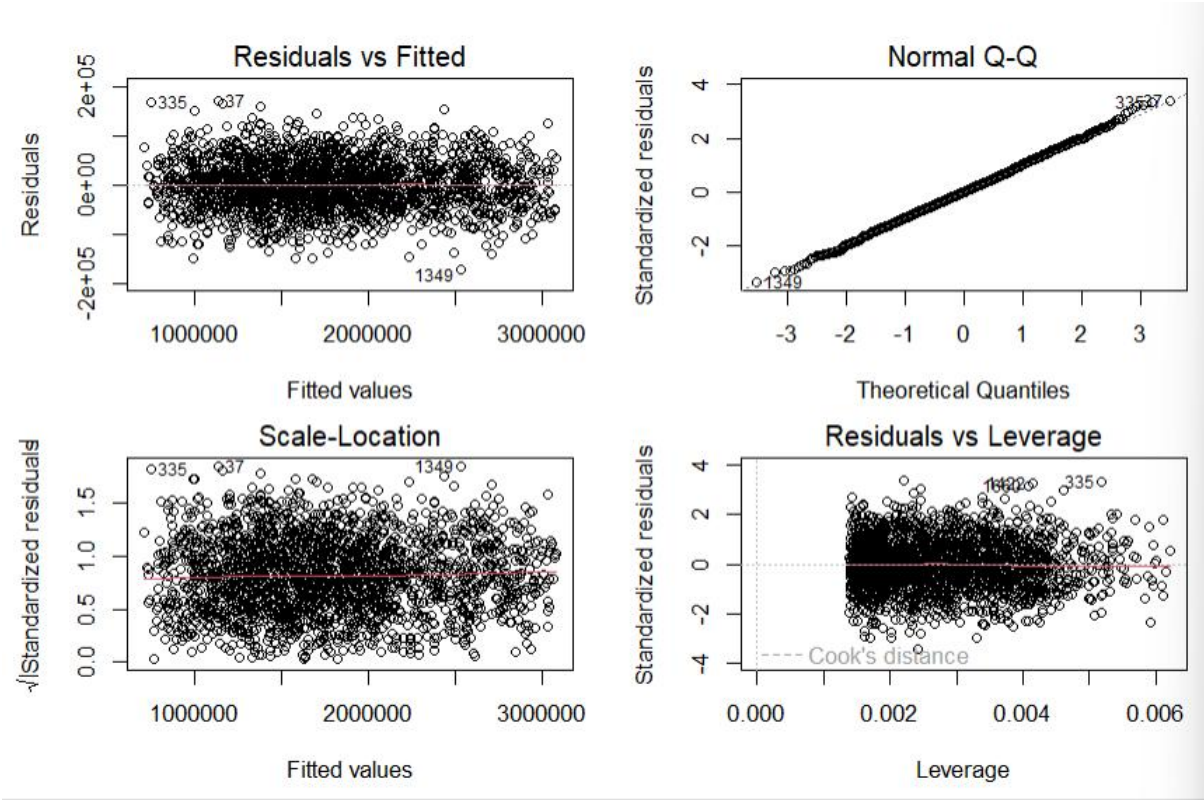
Step 6. Diagnostic Plots

```{r Diagnostic Plots}
  # ---- Diagnostic Plots ----

par(mar = c(4, 4, 2, 2))
par(mfrow = c(2, 2))
plot(model)
par(mfrow = c(1, 1))

```

This is just the code the interesting part is the results.



Diagnostic Plots Analysis Report

1. Residuals vs Fitted

- Random scatter around zero with no clear pattern
- Confirms **linearity assumption is met**
- The few labeled points (335, 37, 1349) are just the most extreme residuals but nothing alarming

2. Normal Q-Q

- Points follow the diagonal line almost perfectly
- Confirms **residuals are normally distributed**
- Slight deviation at the tails is completely normal

3. Scale-Location

- Red line is essentially flat
- Random scatter with consistent spread
- Confirms **homoscedasticity — equal variance across all fitted values**

4. Residuals vs Leverage

- No points outside the **Cook's distance dashed line**
- Confirms **no influential outliers** affecting the model

- All points have low leverage

Overall conclusion — all four model assumptions are satisfied:

- Linearity
- Normality of residuals
- Homoscedasticity
- No influential outliers